ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

SENSORS AND INSTRUMENTATION

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 3rd year Sensors and Instrumentation paper 1 · practice a.

Written by Academic Central for practice, following sensors, actuators and instrumentation content in South African mechatronics modules (e.g. University of Pretoria MBB 410 Mechatronics: sensors and actuators, signal conditioning, sampling and anti-aliasing, digital communication and discrete-time control). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct reasoning in the descriptive answers and answers that differ only through rounding.

QUESTION 1

1.1 Accuracy: how close a reading is to the true value ✓ (6)
Precision: how close repeated readings are to each other ✓
Resolution: the smallest change in input that can be detected ✓
Sensitivity: the change in output per unit change in input ✓
Range: the lowest to the highest input it can measure ✓
Hysteresis: a different output for the same input, depending on whether the input is rising or falling ✓

1.2.1 S = (1.5 − 0.5) / (100 − 0) ✓ = 10 mV/°C ✓ (2)

1.2.2 T = (1.12 − 0.5) / 0.01 = 62 °C ✓ (1)

1.3 Systematic: the same error every time in the same direction (e.g. a zero offset), reduced by calibration ✓ (2)
Random: varies unpredictably from reading to reading (noise), reduced by averaging many readings and filtering ✓

1.4 Mean = 51.5 / 5 = 10.3 mm ✓ (4)
Squared deviations: 0.01, 0.01, 0.04, 0, 0.04, sum 0.10 ✓
$s = \sqrt{(0.10 / 4)}$ ✓ = 0.16 mm ✓

1.5.1 $T(t) = 100 − 80\ e^{(-t/\tau)}$ ✓ (3)
$T(10) = 100 − 80\ e^{-2}$ ✓ = 89.17 °C ✓

1.5.2 About $4\tau = 20$ s ✓ (2)
A slow sensor reports old values; in a control loop this delay (lag) can make the response slow or oscillating ✓

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QUESTION 2

2.1.1 $\Delta R = GF \times \varepsilon \times R = 2 \times 500 \times 10^{-6} \times 120$ ✓ = 0.12 Ω ✓ (2)

2.1.2 $V_{out} \approx V \times GF \times \varepsilon / 4$ ✓ $= 5 \times 2 \times 500 \times 10^{-6} / 4$ ✓ = 1.25 mV ✓ (3)

2.1.3 The bridge turns the very small change in resistance into a small voltage around zero that can be amplified ✓ (2)
A dummy gauge in the next arm, on the same material but not strained, changes with temperature in the same way and cancels the temperature effect ✓

2.2 Thermocouple: widest range (to over 1 000 °C), rugged, low accuracy, needs cold-junction compensation ✓ (3)
RTD (e.g. PT100): very accurate, stable and almost linear, moderate range ✓
Thermistor: very sensitive and cheap but very non-linear, small range ✓

2.3 R = 100(1 + 0.003 85 × 80) ✓ = 130.80 Ω ✓ (2)

2.4 θ = (2 / 5) × 300° ✓ = 120° ✓ (2)

2.5 Counts per revolution = 1 024 × 4 = 4 096 ✓ (3)
2 048 / 4 096 = 0.5 revolution in 0.1 s = 5 r/s ✓
5 × 60 = 300 r/min ✓

2.6 Ultrasonic: cheap, works on glass and dark objects, but slow, with a wide beam and poor angular resolution ✓ (3)
Infrared: cheap and small, but short range and affected by sunlight and surface colour ✓
LiDAR: accurate, long range and a fast, narrow beam (scans give a map), but more expensive ✓

2.7 The gyroscope gives a smooth, fast angle by integrating the rate, but drifts over time ✓ (3)
The accelerometer gives the angle from gravity without drift, but is noisy and upset by vibration and acceleration ✓
Combining them (complementary or Kalman filter) gives a fast, accurate angle without drift ✓

2.8 Inductive: detects only metal objects, using a changing magnetic field (eddy currents) ✓ (2)
Capacitive: detects almost any material, including plastics, liquids and powders, by a change in capacitance ✓

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QUESTION 3

3.1 (a) DC motor: high speed and power, simple speed control ✓ (3)
(b) Stepper motor: accurate positioning in open loop, without an encoder ✓
(c) Servo motor: built-in position control to a set angle ✓

3.2 200 full steps × 16 = 3 200 microsteps per revolution ✓ (3)
2 mm / 3 200 ✓ = 0.000 625 mm (0.63 µm) per microstep ✓

3.3 Advantage (any one): fast, simple, cheap, high force for their size, safe in explosive areas ✓ (2)
Disadvantage (any one): hard to control position accurately, needs a compressor and air lines, noisy, air is compressible ✓

3.4 Speed = 3 000 / 20 = 150 r/min ✓ (2)
Torque = 0.2 × 20 × 0.9 = 3.6 N·m ✓

3.5 $P_{out} = mgv = 20 \times 9.81 \times 0.05$ ✓ = 9.81 W ✓ (5)
$P_{in} = 9.81 / 0.7 = 14.01$ W ✓
I = 14.01 / 24 ✓ = 0.58 A ✓

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QUESTION 4

4.1.1 G = 2.5 V / 1.25 mV ✓ = 2 000 ✓ (2)

4.1.2 $2\,000 = 1 + 50\,000 / R_G$ ✓ (2)
$R_G = 50\,000 / 1\,999 = 25.01\ \Omega$ ✓

4.2 Choose f_c above 100 Hz but well below f_s/2 = 500 Hz, e.g. about 160 Hz ✓ (3)
R = 1 / (2π f_c C) = 1 / (2π × 160 × 100 × 10^{-9}) ≈ 10 kΩ ✓
With R = 10 kΩ: f_c = 1 / (2π × 10^{4} × 10^{-7}) = 159.15 Hz ✓
(Accept any f_c between 100 Hz and 500 Hz with the correct calculation.)

4.3.1 Any TWO: the current is the same everywhere in the loop, so cable resistance does not cause errors; it is less affected by noise than a voltage signal; 0 mA shows a broken wire (live zero at 4 mA); the loop can power the sensor ✓✓ (2)

4.3.2 P = (12 − 4) / 16 × 10 = 5 bar ✓ (1)

4.3.3 R = 5 V / 20 mA = 250 Ω ✓ V = 12 mA × 250 Ω = 3 V ✓ (2)

4.4 Any THREE: use shielded cable with the shield earthed at ONE end; use twisted-pair wires; keep sensor cables away from motor and power cables; use differential inputs; avoid ground loops by earthing at one point; filter the signal ✓✓✓ (3)

4.5 a = (5 − 0) / (2 − (−2)) ✓ = 1.25 ✓ (3)
b = 2.5 V (so −2 V gives 0 V and +2 V gives 5 V) ✓

4.6 SNR = 20 log(2 / 0.002) ✓ = 60 dB ✓ (2)

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QUESTION 5

5.1 Resolution = 3.3 / 4 096 ✓ = 0.81 mV ✓ (3)
Maximum quantisation error = ±½ LSB = ±0.40 mV ✓

5.2 250 Hz is above f_s/2 = 200 Hz, so it is aliased ✓ (3)
Alias frequency = |f − f_s| = |250 − 400| ✓ = 150 Hz ✓

5.3.1 4th output: (10 + 12 + 30 + 11) / 4 ✓ = 15.75 cm ✓ (3)
5th output: (12 + 30 + 11 + 13) / 4 = 16.50 cm ✓

5.3.2 The 30 cm reading is a single spike (outlier); a moving average spreads it into the next outputs ✓, while a median filter ignores it completely ✓ (2)

5.4 u[k] = Kp e[k] + Ki T Σ e[i] + Kd (e[k] − e[k−1]) / T ✓✓ (4)
The P term reacts to the present error; the I term adds up the error over time (sum × T) to remove the steady-state error ✓; the D term uses the change since the last sample to add damping ✓

5.5 Any ONE, explained: CAN is designed for long cables (tens of metres) with differential signals that resist noise, while I2C is meant for short distances on one circuit board ✓✓; CAN has error checking and message priority ✓ (3)

5.6 θ = 0.98(10 + 20 × 0.01) + 0.02 × 12 ✓ = 0.98 × 10.2 + 0.24 = 10.24° ✓ (2)

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TOTAL: 100

END OF MARKING GUIDELINE